

Table 5. Controls and selected grid extrema

Panel A. Scenario identity

Summary type	Source grid	Scenario id	Control label	Initial state definition
control	figure07_controls	baseline_sensitivity	Baseline sensitivity	unpolarized I6/6
control	figure07_controls	zero_mixing	Zero mixing	unpolarized I6/6
control	figure07_controls	spin_independent_reaction	Spin-independent reaction	unpolarized I6/6
control	figure07_controls	fast_relaxation	Fast relaxation	unpolarized I6/6
control	figure07_controls	rapid_escape	Rapid escape	unpolarized I6/6
control	figure07_controls	doublet_benchmark	Doublet benchmark	doublet-manifold mixture PD/2
control	figure07_controls	quartet_benchmark	Quartet benchmark	quartet-manifold mixture PQ/4
selected grid minimum	figure04_mixing_escape_heatmaps	mix_escape_kqkd_0	Not available	unpolarized I6/6
selected grid maximum	figure04_mixing_escape_heatmaps	mix_escape_kqkd_10	Not available	unpolarized I6/6
selected grid minimum	figure05_mixing_relaxation_heatmaps	mix_relax_kqkd_0	Not available	unpolarized I6/6
selected grid maximum	figure05_mixing_relaxation_heatmaps	mix_relax_kqkd_10	Not available	unpolarized I6/6
selected grid minimum	figure06_reaction_selectivity	selectivity_quartet-manifold_mixture	Not available	quartet-manifold mixture PQ/4
selected grid maximum	figure06_reaction_selectivity	selectivity_quartet-manifold_mixture	Not available	quartet-manifold mixture PQ/4

The extrema describe the sampled computational grid only. Presentation values are rounded. All 13 source rows and all 53 source columns remain in table05_controls_and_extrema.csv; 41 provenance/support columns are source-only. 'Not available' is distinct from numeric zero.

Table 5. Controls and selected grid extrema

Panel B. Coordinates and results

Summary type	Mixing over reference	Radical relaxation over reference	Escape over reference	kQ divided by kD	Primary superoxide (radical eq. per encounter)	Escape yield (per encounter)	Unresolved probability
control	1	0.1	1	0.1	0.242129	0.757828	0.000043
control	0	0.1	1	0.1	0.233861	0.766067	0.000072
control	1	0.1	1	1	0.500000	0.500000	1.125e-07
control	1	100	1	0.1	0.285255	0.714731	0.000014
control	1	0.1	100	0.1	0.003966	0.996034	0
control	1	0.1	1	0.1	0.445496	0.554487	0.000017
control	1	0.1	1	0.1	0.140445	0.859498	0.000057
selected grid minimum	0	0.1	100	0	0.003300	0.996700	0
selected grid maximum	15.849	0.1	0	10	1.000000	0	2.514e-15
selected grid minimum	0	0	1	0	0.166667	0.833110	0.000224
selected grid maximum	100	100	1	10	0.873791	0.126209	0 (roundoff)
selected grid minimum	0	0.1	10	0	0.000595	0.999405	0 (roundoff)
selected grid maximum	0	0.1	0.1	10	0.989625	0.010375	3.960e-07

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